

ECONOMICS MEANING AND BASIC COMCEPT

HISTORY OF ECONOMICS

The subject **economics** has no specific definition. It has been defined in many ways by various economists as a **social science** which studies human beings and their behavior. Some of the definitions given by some of the experts in the subject include the following:

Alfred marshal defined economics as “a study of mankind in the ordinary business of life.” This definition simply emphasizes that economics has something to do with the study of human beings in relation to their daily economic activities.

Adam smith saw it as “an inquiry into the nature and causes of wealth of nations.” To him economics is all about the making of wealth. Adam smith was the first person to put a work human together on economics by writing a **treatise**, “wealth of nation”. He is popularly regarded as the **father of economics**.

(lord) lionel c. Robbins. He defined economics as “the science which studies **human behavior** as a relationship between ends and **scarce means** which have **alternative uses**”. This definition is all-embracing because it covers some major aspects of economics such as scarcity, wants, human behavior and choice. The **ends** in the definition refer to human wants, desires or needs. Human wants are numerous or many, relative to available resources required to satisfy them. The scarce means refers to the limited available resources used in satisfying the numerous human wants. In other words, the resources required to satisfy human wants which are referred to as the ‘means’ are scarce or not many, relative to their demand. The alternative uses in the definition means that these scarce resources can be used for different purposes. In other words, the more pressing needs have to be satisfied first, leaving others that are less important.

NATURE AND SCOPE OF ECONOMICS

Economics belongs to a group of subject called **social science**. Other social science subjects include sociology, psychology, government, political, religious studies anthropology, philosophy, etc. Economics is ne regarded as a social science because it studies human behavior. For example, if the price of a commodity rises, people will buy less, while if the price of such commodity decreases, people will buy more, all things being equal. Economics is also concerned with human behavior, such as how people achieve their wants. Man interacts with people in the process of buying and selling. Economics as a social science subject 1s also concerned with the study of firms or companies and the government which is responsible for the provision of goods and services for its people in order to satisfy their wants.

BASIC CONCEPTS OF ECONOMICS

The basic concepts or elements of economics are: wants, scarcity, scale of preference, choice and
Opportunity cost,

WANTS

Definition: want may be defined as insatiable desire or need by human beings to own goods or services that give satisfaction. The basic needs of man include: food, housing and clothing. Human needs are many. They include tangible goods like houses, cars, chairs, television set, radio, etc.

SCARCITY

Definition: scarcity is defined as the limited supply of resources which are used for the satisfaction of unlimited wants. In other words, scarcity is the inability of human beings to provide themselves with all the things they desire or want.

SCALE OF PREFERENCE

Scale of preference is defined as a list of unsatisfied want arranged in the order of their relative importance. In other words, it is a list showing the order in which we want to satisfy our wants arranged in order of priority. In the scale of preference, the most pressing wants come first and the least pressing ones come last. It is first in the least has been satisfied that there will be room for the satisfaction of the next.

CHOICE

Definition: choice can be defined as a system of selecting or choosing one out of a number of alternatives. Human wants are many and we cannot satisfy all of them because of our limited resources. We, therefore, decide which of the wants we can satisfy first. Choice arises as a result of numerous human wants and the scarcity of the resources used in satisfying these wants.

OPPORTUNITY COST

Definition: opportunity cost is defined as an expression of cost in terms of forgone alternatives. It is the satisfaction of one's want at the expense of another want. It refers to the wants that are left unsatisfied in order to satisfy another more pressing need. Human wants are many, while the means of satisfying them are scarce or limited. We are therefore, faced with a problem where we have to choose one from a whole set of human wants, to choose one means to forgo the others.

WHY WE STUDY ECONOMICS

We study economics for the following reasons:

- (1) **Allocation of resources:** The study of economics enables the government to allocate scarce resources to various sectors of the economy.
- (2) **Development of programmes:** It also enables the government to develop certain programmes that are beneficial to the people.
- (3) **Rational decision:** Economics enables the individuals to choose certain wants among the numerous needs, using their scarce resources.
- (4) **Preparation of budget:** Economics assists the government to determine the expected income and expenditure of a country
- (5) **Solutions to economic problems:** Economics also enables individuals, firms and governments to solve their problems using various principles of the subject.
- (6) **Production:** The study of economics assists us to determine what to produce, when to produce, factors of production and how to produce goods and services required to satisfy human want
- (7) **Provision of basic tools:** The study of economics provides basic tools for analyzing economic problems among individuals, firms and governments.
- (8) **Maximization of profits:** Economics enables traders and businessmen to maximize their profits using economic principles in their business.
- (9) **Consumption of commodities**

Economics also assists us to determine the pattern of consumption of goods and services in our local environment.

(10) Satisfaction of wants: The study of economics helps us to utilize the principles of choice, opportunity cost, scale of preference, etc. in order to satisfy the human wants.

(11) Participation in government: The or age Understanding of economics helps individuals to participate actively in the art of governance.

MICRO-ECONOMICS

Definition: micro-economics refers to the branch of economics which deals with smaller units or components of the economy. It is concerned with the analysis of the basic decision making components of households, individuals, firms and governments. It relates to cost, output, production, pricing and marketing activities of households, firms and governments.

Advantages of micro-economics

(1) Better understanding

Micro-economics helps in the better understanding of the functioning of the various units or components of the economy.

(2) Making of policies

It also enables us to make and develop better policies that will improve the welfare of the people.

(3) Knowledge of vibrant sector

The knowledge of micro-economics enables us to determine the vibrant sector of the economy.

(4) Development of economic tools

The study of micro-economics helps us to develop sound economic tools used for interpreting or solving economic problems.

Disadvantage of micro-economics

(i) Unreliability of data

Data derived from the use of micro-economic principles are not always reliable for the same components or units.

MACRO-ECONOMICS

Definition: macro-economic refers to the branch of economics which deals with larger units or aggregate of the economy. Macro-economic relates to large aggregates such as national income, inflation, unemployment, balance of payment, etc. In summary, macro-economic deals with the broad aggregates in the economy

Advantages of macro-economics

(1) Even distribution of income

The study of macro-economics' helps to make sure that the wealth of the nation is not concentrated in the hands of few individuals or to certain sectors of the economy.

(2) Full employment

In a macro-economy, full employment is provided to a larger proportion of the population.

(3) Provision of goods and services

Goods and services are generally provided for the people in a macro-economic system.

(4) Balance of payment

Macro-economic also ensures adequate balance of payment in the total economy.

(5) Monetary policies

Monetary policies are being completely analyzed and comprehended easily through the study of macro-economics.

(6) Increase in gross domestic product (GDP)

The study of macro-economic has resulted in the increase in gross domestic product (GDP) leading to economic development.

(7) Stability of price

Price stability is obtained in macro-economic by minimizing price disturbances in the economy.

(8) Deficiency in aggregate demand

The causes of deficiency in aggregate demand are made known through the study of macro-economics.

Disadvantages of macro-economics

(i) Statistical difficulties

The study of macro-economic makes statistical data difficult to analyze as a result of the grouping of macro-economic variables.

(ii) Negative grouping of data

In the analysis of macro-economics, data are being negatively grouped to be the same without considering the particular nature of each component of the economy.

OTHER BRANCHES OF ECONOMICS

(1) pure economics

This is concerned with the study of the laws and theories derived from the study of economic behavior,

(2) Applied economics

This is concerned with the application of the laws and theories in analyzing and solving economic problems.

(3) Mathematical economics

This is concerned with the collection and analysis of data as well as statistics.

(4) Monetary economics

This involves the study of money and banking.

(5) Business economics

Business economics is concerned with the study of trade, business organization and Accounting

(6) Development economics

This is concerned with the study of economic planning and national economics

TABLES

Definition: a table is defined as a systematic and orderly arrangement of information, facts or data using rows and columns for presentation, which makes it easier for better understanding. Tables serve as the most commonly used tools for economic analysis.

Example of a table

Let us consider table 2.1. It represents the quantity of motor cars demanded in a country in relation to the price.

Table 2.1: number of vehicles demanded in a country

Price (₦)	No of vehicles demanded
50,000	800
200,000	500
450,000	300
800,000	120
1,000,000	60

Interpretation of the table

From table 2.1, it can be seen that the number of vehicles demanded at a given price varies. When the price was cheap (₦50,000.00), more of the vehicles were in high demand (800 vehicles) but

When the price was high (₦1,000,000.00) only few of the vehicles (60 vehicles) were demanded,

Characteristics of a table

- (1) A table must be very simple.
- (2) It must be easy to understand.
- (3) It must have a title or heading.
- (4) The units of measurement used in the table must be stated.
- (5) The purpose of constructing the table must be stated.
- (6) The sub-headings for the columns and rows used must be stated.
- (7) A table must be numbered if they are many.
- (8) The source of information should be properly stated if it is known.

Importance of tables

- (1) A table helps to summarize the data presented.
- (2) It assists in an orderly arrangement of data.
- (3) A table eases comparison between different classes of data.
- (4) A table helps to bring out the important features in the data.
- (5) A table helps to understand data.
- (6) It makes it easier and faster to make decision through the use of table.

GRAPHS

Definition: graph may be defined as a diagram showing a functional relationship between two variables. Graph is one of the basic tools used by economists for economic analysis. Information

Presented on tables can be translated into graph for better understanding. Examples of graphs are Line graphs, pie graphs (or charts), bar graphs, pictographs (or pictograms)

Features of graph

- (1) All graphs must have a title.
- (2) Graphs must possess appropriate scales.
- (3) All graphs should have x-axis on the horizontal side and y-axis on the vertical side.
- (4) Graphs must be labelled properly for better understanding.

- (5) Graphs must show the source of the data presented.
- (6) The unit of measurement must be indicated in a good graph.
- (7) The x and y axes of the graph must carry different variables,

Importance of graph

- (1) Graphs help to show relationship between two variables.
- (2) Graphs make clearer and quicker impression about quantitative information illustrated in tabular form.
- (3) Graphs also help to interpret value of variables.
- (4) Graphs provide basis for comparing variables provided in the table.
- (5) Graphs make it possible for changes in variables on quantities to be expressed.
- (6) Values or quantities provided in the table are better understood and appreciated either in graphs or diagrams.

LINE GRAPH

Definition: line graph, as the name implies, is a line used for data where emphasis is on continuous change. In other words, line graph is used to join or connect the highest and the lowest points of a group of data. Line graph could be straight or curve.

Example

The data in table 2.2 represent the production of cocoa by Adamawa state between 2011 and 2021, represent the information by a line graph.

Table 2.2: Cocoa production by Adamawa State from 2011 to 2021

Year	Cocoa production (Tonnes)
2011	8
2012	10
2013	12
2014	14
2015	16
2016	18
2017	20
2018	22
2019	24
2020	26
2021	28

Solution

Fig. 2.1 graph showing cocoa production (in Tonnes) by Adamawa state between 2011 and 2021. The graph in fig. 2.1 is a good example of a straight line graph.

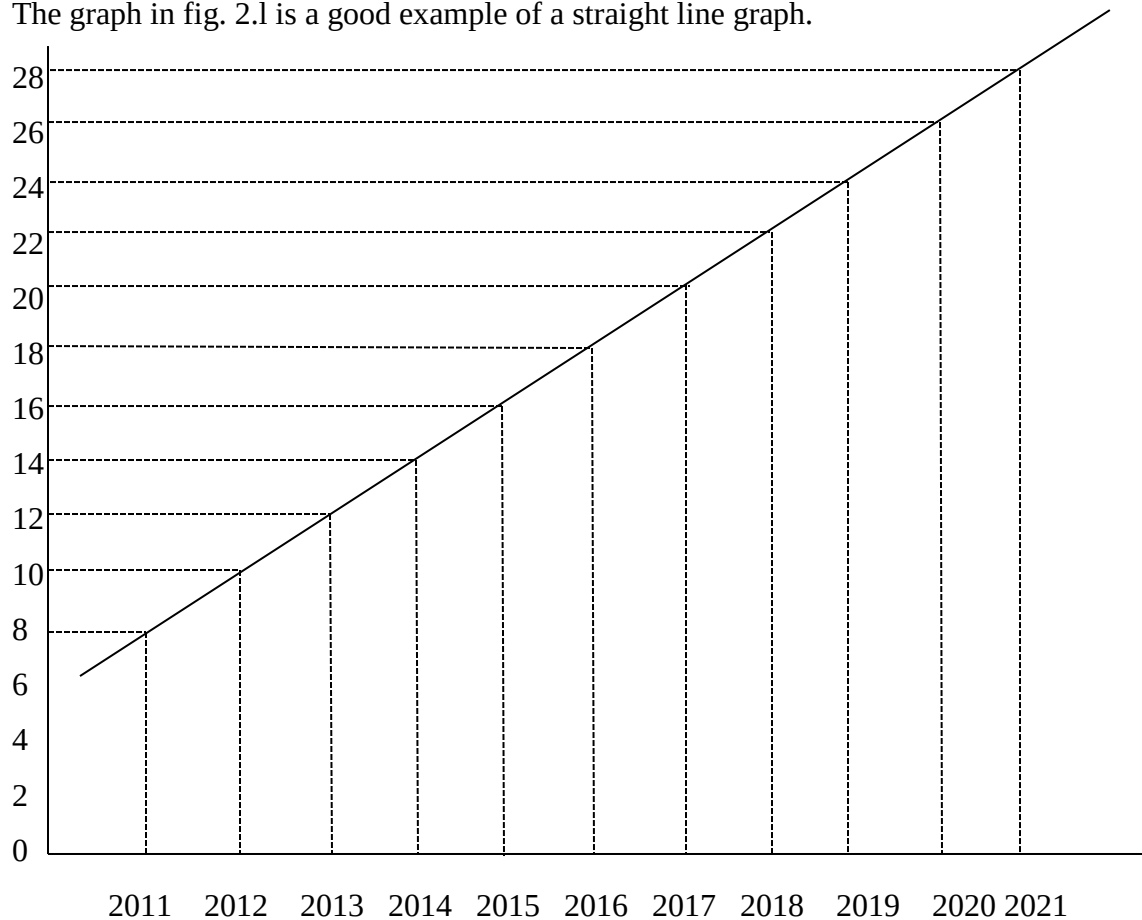


Fig. 2.1: cocoa production by Adamawa state from 2011 to 2021

PIE CHARTS [OR GRAPHS]

Definition: a pie chart or graph is a simple circle of any convenient size which is divided into sections or sectors, each of which is proportional to the quantity or value it represents.

The pie chart is usually measured either in percentage or in degrees with the aid of a mathematical device called a protractor. The entire circle is represented by 360° or 100% and each sector is measured in degrees with the aid of a protractor.

Methods of construction

- (i) Add up the total figure of the product or export product value in million value under consideration.
- (ii) Work out the percentage or degree of the total which each component part
- (iii) Draw circle of a convenient size.
- (iv) Divide the circle up into sectors subtended by these angles calculated in step ii above using a protractor.
- (v) Write the percentage or degree of each solution sector by it.
- (vi) Use key where the information cannot be written inside the circle to show what each sector represents

Example

Represent the information in table 2.3 by means of pie chart, the value of the Table 2.3: value of the most important exports of Nigeria in 2011

Export product	Value million
Cocoa	200
Groundnut	140
Petroleum	240
Coal	50
Other	90

Solution

- (i) Add up the total value of all the products, i.e. $200 + 140 + 240 + 50 + 90 = 720$.
(ii) Arrange your workings or calculations as follows:

Export product	Value in million (₦)	Workings in percentage (%)	Workings in degree (0°) angle of sector
Cocoa	200	$\frac{200}{720} \times 100 = 29.8\%$	$\frac{200}{720} \times 360 = 100^\circ$
Groundnut	140	$\frac{140}{720} \times 100 = 19.5\%$	$\frac{140}{720} \times 360 = 70^\circ$
Petroleum	240	$\frac{240}{720} \times 100 = 33.3\%$	$\frac{240}{720} \times 360 = 120^\circ$
Coal	50	$\frac{50}{720} \times 100 = 6.9\%$	$\frac{50}{720} \times 360 = 25^\circ$
Others	90	$\frac{90}{720} \times 100 = 12.5\%$	$\frac{90}{720} \times 360 = 45^\circ$
Total	720	100%	360°

- (iii) Using the values in degree, i.e. 100° , 70° , 120° , 25° and 45° , a pie chart in fig. 2.2 is with a pie chart. Show your workings clearly. Now drawn using a protractor.

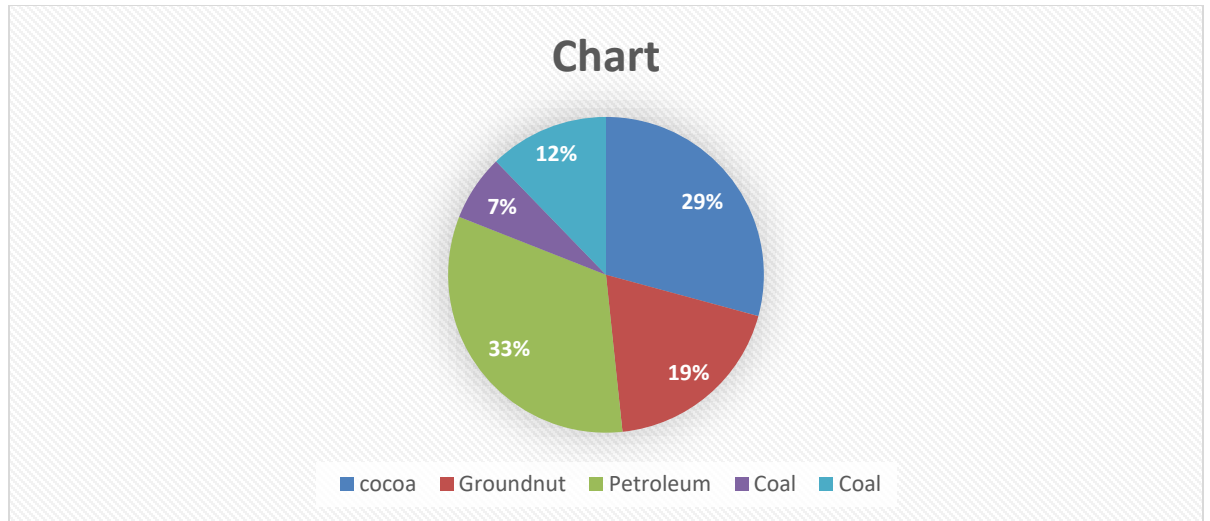


Fig.2.2 pie chart showing value of the most import export of Nigeria in 2011.

BAR CHARTS [OR GRAPHS]

Meaning: bar chart or graph is a graph made up of bars of rectangles which are of equal width and whose lengths are proportional to the quantities they represent. The major characteristics of the bar chart are that the body of the bars must not touch each other. There must be a space or gap between one bar and another. Bar chart may be arranged vertically or horizontally.

Types of bar charts

There are three major types of bar charts. These are (i) simple bar chart (ii) component bar chart and (iii) multiple bar charts.

- (a) **Simple bar charts:** simple bar chart (fig. 2.4) is used when the data given are made up of only one item or component. The bar chart can be presented by tabulated data with evenly spaced bars, separated by gaps with the length proportional to the magnitude of the value given.

Example

The table in 2.5 represents the quantity of cocoa production in Nigeria between 2015 and 2022. Represent the information or data by a bar chart.

Table 2.4(a): cocoa production in Nigeria between 2015

Year	Cocoa production (tons)
2015	7,000
2016	1,500
2017	3,000
2018	6,000
2019	8,500
2020	5,000
2021	4,000
2022	3,500

Solution

(i) From the figures given, using the scale: 1cm:1000 tonnes, the figures can be reduced to simple ones when they are divided by 1000.

(ii) The new values will now be.

Table 2.5(a): cocoa production in Nigeria between 2015 and 2022.

Year	Cocoa production (tons)
2015	7.0
2016	1.5
2017	3.0
2018	6.0
2019	8.5
2020	5.0
2021	4.0
2022	3.5

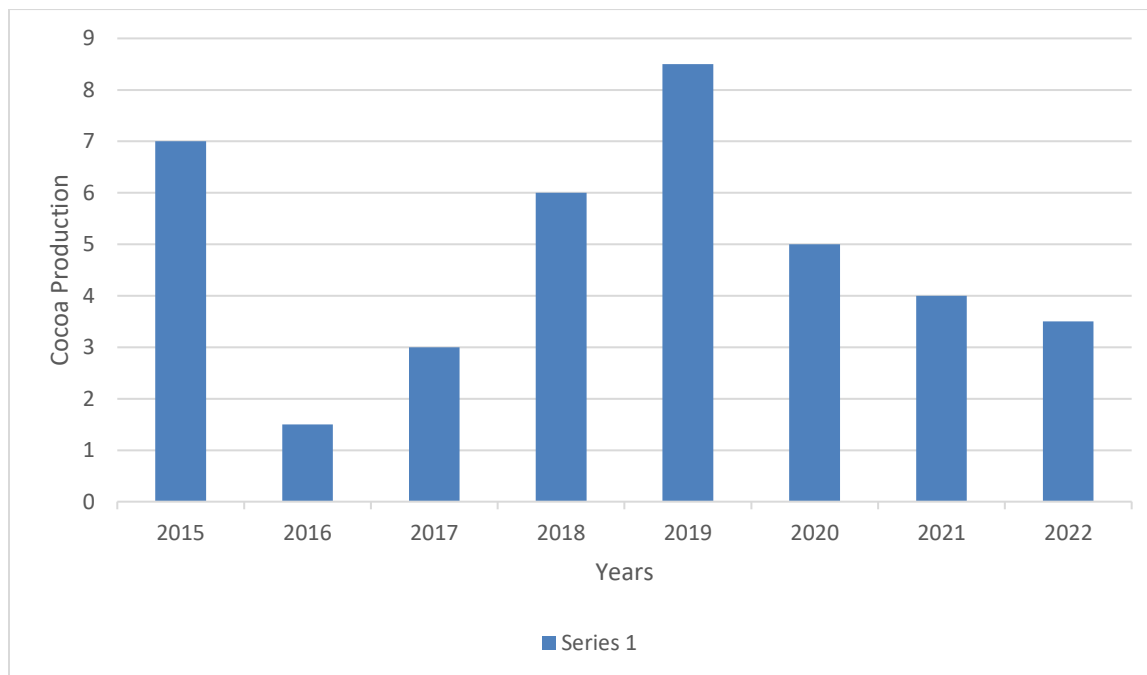


Fig. 2.4; simple bar chart showing cocoa production in Nigeria (2015-2022).

Useful information to be derived from bar graph

- (i) The year with the highest cocoa production can be calculated. The year is 1964.
- (ii) The year with the least cocoa production can also be calculated. The year is 1961.
- (iii) The total cocoa production from 1960 to 1967 can be calculated, i.e. $7,000 + 1,500 + \dots + 4,000 + 3,500 = 38,500$.
- (iv) The average or mean production can also be calculated by adding the total production (as in (iii) above) and dividing it by the number of years (8 years), i.e.

$$\frac{7,000 + 1.5000 + \dots + 4000 + 3500}{8}$$

$$= \frac{38.500}{8} = 4,812.5 \text{ tons/year}$$

(b) Component bar charts: A component bar chart (fig. 2.5) is used when the data involved are of two variables.

Example

Represent the population of males and females (in million) (table 2.6) in some towns in Adamawa state in 1999 by a bar chart.

Table 2.6: population in millions of males and females of some towns in Adamawa state

Towns	Male (m)	Female (f)	Total (m/f)
Mubi	80	60	140
Michika	60	50	110
Song	70	30	100
Madagali	30	15	45

Solution

Using a graph sheet, choose a suitable scale for the graph, e-g. 1cm to represent 10 units, the graph (component bar chart in fig. 2.5) is then drawn.

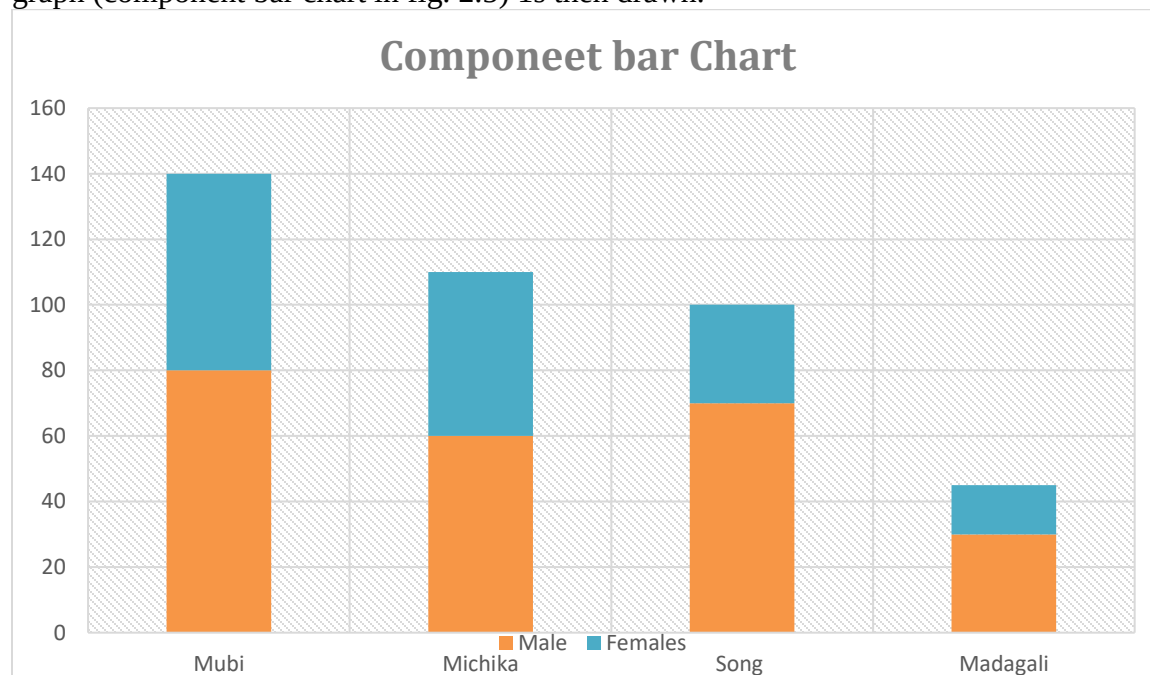


Fig 2.5 component bar chart showing the population of some towns in Adamawa state in 1999

(c) Multiple bar charts: The multiple bar chart (fig. 2.6) is used when there are about three or more variable in a stands for a component variable.

Example

Represent Nigeria's export of cocoa (table 2.7) from mubi, michika and song in 1960,1970 and 1980 by a bar chart.

Table 2.7: Nigeria's export of cocoa from mubi, michika and song in 1960, 1970 and 1980 by bar chart

Towns	1960(tons)	1970(tons)	1980(tons)
Mubi	30	40	50
Michika	40	30	60
Song	50	70	40

Solution

Using a graph sheet, choose a suitable scale for the graph, e.g. Lcm represents 10 units, the graph (multiple bar chart in fig. 2.6) ii than drawn.

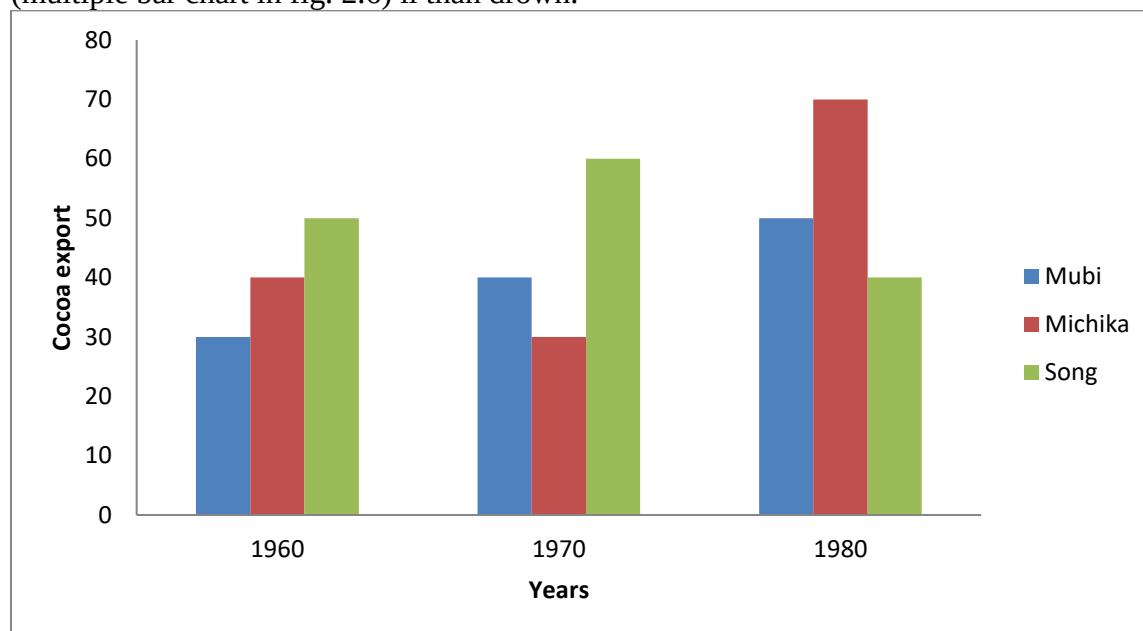


Fig. 2.6: multiple bar chart showing Nigeria's exports of cocoa from Mubi, Michika and Song in 1960, 1970 and 1980.

FREQUENCY DISTRIBUTION

Meaning: frequency distribution refers to the Arrangement of data or information in tabular form to reflect their frequencies. Frequency refers to the number of times a particular event or information is usually used when data presented are large and most of the numbers may appear more than once,

Example

Represent the marks scored by 30 biology students in SSI by frequency distribution using the following data.

20	8	12	4	18	18	18
20	12	6	18	20	4	12
8	18	12	8	8	18	20
2	20	18	18	20	8	2
4	18					

Solution

Arrange the data in the following manner.

Table 2.10: marks scored by 30 biology students in SSI

Score (x)	Tally or Counts	Frequency
2	II	2
4	III	3
6	I	1
8	HHH	5
12	IIII	4
18	HHH-IIII	9
20	HH I	6
		30

MEASURE OF CENTRAL TENDENCY

Meaning: measure of central tendency, also called measure of location, is the statistical information that gives the middle or center or Average of a set of data. They are all regarded as forms of averages.

There are five measures of central tendency they are:

- (i) Arithmetic mean
- (ii) Median
- (iii) Mode
- (iv) Geometric mean
- (v) Harmonic mean

Arithmetic mean, median and mode will be discussed now, while geometric mean and harmonic Mean will be discussed later.

THE ARITHMETIC MEAN

Definition: The arithmetic mean, also popularly referred to as the “mean”, is the average of a series of figures or values. It is obtained by dividing the sum of these figures by the total number of the figures or values. It is also the average of a collection of observation. The arithmetic mean is the most popularly used measure of central tendency.

Formula for calculating arithmetic mean

$$\text{Arithmetic mean, } = \bar{x} = \frac{\sum fx}{n}$$

Where $\bar{x}$ = arithmetic mean

$\sum$ = represents a Greek letter denoting

“Sum of”

X = series of figures in a given data

$\sum x$ = the total of the values of series of figures in a given data

n = number of figures or elements

Note: This formula is used especially when the figures are small and ungrouped.

Example 1

Calculate the arithmetic mean for the scores of eight students in NECO economics examination in the year 1999. The scores are: 14, 18, 24, 16, 30, 12, 20 and 10.

Solution

Step I: Add up the numbers or scores $\sum x = 14 + 18 + 24 + 16 + 30 + 12 + 20 + 10 = 144$

Step II: N = no. Of figures or scores which is = 8

Step III: $\bar{x} = \frac{\sum x}{n} = \frac{144}{8} = 18$

Note: when the figures given in a data are large and in most cases repeat themselves, then frequencies are used. Frequency is the number of times a particular event or information occurs. Frequency distribution is usually used when data presented are large and most of the numbers appear more than once.

In this case the formula for calculating the arithmetic mean will change slightly to:

Arithmetic mean, $\bar{x} =$

Where f no. of times a particular number occurs (frequency). Other symbols remain the same.

Example 2

Calculate the mean of the following sets of numbers:

8, 16, 24, 8, 12, 12, 18, 24, 10, 16, 20, 24, 24, 12, 24, 12, 16, 24, 18, 18.

Solution

Step I: identify the numbers that occur in the set, i.e. 8, 10, 12, 16, 18, 20 and 24. Arrange these numbers in a frequency distribution table (table 2.11).

Step II: arrange the numbers starting from the smallest number, which is 8, to the highest number, which is 24, as shown in table 2.11.

Step III: Arrange the figures or numbers in a frequency distribution table as shown in table 2.11.

table 2.11: frequency distribution

Numbers (x)	Tally or Counts	Frequency (f)
10	II	2
12	I	1
16	IIII	4
18	III	3
20	III	3
24	I	1
	HH I	6

$$n = 20$$

Step IV: apply the formula arithmetic mean, $\bar{x} = \frac{\sum fx}{n}$

$$\frac{(8 \times 2) + (10 \times 1) + (12 \times 4) + (16 \times 3) + (18 \times 3) + (20 \times 3) + (24 \times 1)}{20}$$

$$= \frac{16 + 10 + 48 + 48 + 54 + 60 + 24}{20}$$

$$= \frac{340}{20} = 17.0$$

Mean of group data

The arithmetic mean can also be prepared for grouped data. In this case the class mark (mid-Points) of the individual class interval is used for the X-column.

Formula used is Arithmetic mean $\bar{X} = \frac{\sum fx}{\sum f}$

Example 3

Calculate the mean of the following marks scored by students in an economics examination:

8, 31, 45, 38, 22, 28, 16, 51, 65, 48, 6, 24, 18, 12, 16, 48, 38, 50, 44, 6, 18, 16, 24, 32, 36, 26, 14, 20, 12, 18.

Solution

- (i) Use the steps as in example 2
- (ii) Use a class interval of 0 - 9, 10 - 19, 20 - 29, etc. As shown in table 2.12.
- (iii) prepare a frequency table as in table 2.12

table 2.12: frequency table for marks scored by students in economics examination.

Scores (Groupings)	X Class marks	tally or Event	f Frequency	fx
0-9	4.5	III	3	13.5
10-19	14.5	HHH IIII	9	130.5
20-29	24.5	HHH I	6	147.0
30-39	34.5	HHH	5	172.5
40-49	44.5	IIII	4	178.0
50-59	54.5	II	2	109.0
60-69	64.5	I	1	64.5
			$\sum f = 30$	$\sum fx = 815$

$$\text{Arithmetic mean } x = \frac{\sum fx}{\sum f}$$

$$= \frac{815}{30} = 27.2$$

Advantages of arithmetic means

- (i) Arithmetic mean is very easy to calculate
- (ii) It gives an exact value
- (iii) it is the best known average.
- (iv) it is very easy to understand.
- (v) It provides a good method of comparing values.
- (vi) it makes use of all available information in a data.

Disadvantages of arithmetic means

- (i) Arithmetic means cannot be obtained graphically.
- (ii) Certain facts in arithmetic mean may not be revealed.
- (iii) it may be difficult to obtain without calculations.
- (iv) it can lead to distorted result.
- (vii) It may be badly affected by extreme values in a distribution.

THE MEDIAN

Definition: the median is defined as an average, which is the middle value when figures are arranged in order of magnitude. In an even distribution, the median is the average of the two middle numbers. In other words, the median of a distribution is the middle value when the observations are arranged in order of magnitude starting with either the smallest or the largest number. The median is therefore the value of the middle item.

How to calculate the median?

A. when the numbers involved are odd numbers, the median will be the middle number.

Example 1

Find the median of the following sets of Values: 2, 8, 11, 13, 15, 6, 9, 20, 7

Solution

Formula: median $\frac{(N+1)}{2}$

Where n is the number of items (observation). This formula is applicable only when n is odd.

Step 1 First arrange the numbers in ascending or descending magnitude

(a) In ascending magnitude, we have 2, 6, 7, 8, 9, 11, 13, 15, 20.

Count the numbers of values involved.

There are nine (9) numbers involved (**odd number**)

Since there are 9 numbers involved, the middle number is the 5th number and the 5th number is 9. Therefore, the median = 9.

Alternatively, the formula can be used

$$\begin{aligned}\text{Median} &= \frac{(N + 1)}{2} = 9 \\ &= \frac{9 + 1}{2} = \frac{10}{2} = 5 \\ &= 5^{\text{th}} \text{ number}\end{aligned}$$

The 5th number in ascending magnitude is 9 Therefore the median = 9

(b) in descending magnitude, we have 20,

15, 13, 11, 9, 8, 7, 6, 2

Since there are nine (9) numbers involved, the 5th number, which is the middle one (9), is the Median. Therefore, the median = 9

(B) When even numbers are representing the number of events in a data, the two middle values are taken; add them and divide them by two. The median will be the arithmetic mean of the two middle numbers.

Example 2

Find the median of the following numbers: 20, 8, 12, 8, 10, 14, 18, 5

Solution

Step 1 First arrange the numbers in ascending or descending magnitude. In ascending magnitude, we have: 5, 8, 8, 10, 12, 14, 18, 20

Step II count the number of values involved there are eight (8) values (even numbers).

Step III Since there are eight (8) numbers Involved, the middle will be the 4th and the 5th numbers, which are 10 and 12.

Step IV To get the median, then add 10 + 12 Together and divide them by 2

$$\text{I.e. } \frac{10+12}{2} = \frac{22-11}{2} = 11$$

The median = 11

Alternatively, the formula can be used

$$\text{Median} = \frac{(n+1)}{2}$$

Where n = 8

$$\frac{(8+1)}{2} = \frac{9}{2} = 4.5$$

Therefore, the median is found between the 4th And 5th number, i.e. Between 10 and 12. The middle between 10 and 12 is 11

Therefore, median = 11

(c)when a group data is involved, cumulative frequency is used

This is used when items or values are large and arranging them in ascending order may not work.

The formula will now be:

Member for odd

(i) Median $(\frac{N+1}{2})^{\text{th}}$ number of items number of items, i.e. Where N is even

(ii) Median = $\frac{(\frac{N+1}{2})^{\text{th}} + (\frac{N+1}{2})^{\text{th}}}{2}$ Number for Even number of Items, i.e. Where N is even

Where N is the summation of all the frequency and this is the terminal cumulative frequency.

Example 3

Use the information in table 2.13 and calculate the median age of some SSII students.

Table 2.13: age distribution of SSII students.

Age (yrs)	6	8	10	11	12	13	14
Frequency	5	10	3	8	7	10	8

Solution

A cumulative frequency table (table 2.14) is prepared for the distribution.

Table 2.14: cumulative frequency for age distribution of SSII students.

Age (yrs)	6	8	10	11	12	13	14
No. of student (frequency)	5	10	3	8	7	10	8
Cumulative Frequency	5	15	18	26	33	43	51

From table 2.14, there are 51 members as indicated by the terminal (last) cumulative frequency.

Since the members are **odd (51)** the median age will be $\frac{(N+1)}{2}^{\text{th}}$ member

$$\text{Median age} = \frac{(51+1)}{2}^{\text{th}}$$

$$= \frac{52}{2}$$

$$= 26^{\text{th}} \text{ member}$$

The 26th falls within the cumulative under the age of 11 years in the table above. Therefore, the median age = 11 years

Example 4

The data in table 2.15 represents the marks scored by government students in NECO examinations. Calculate the median score.

Table 2.15: marks scored by government students in NECO examinations.

Marks %	12	18	24	30	36	40	48
Frequency	6	1	10	8	12	3	4

Solution

A cumulative frequency table (table 2.16) is prepared for the distribution.

Table 2.16: cumulative frequency table for marks scored by government students in NECO examination.

Marks %	12	18	24	30	36	40	48
Frequency	6	1	10	8	12	3	4
Cumulative Frequency	6	7	17	25	37	40	44

From the table (table 2.16), there are 44 members as indicated by the terminal (last) Cumulative frequency.

Since this **44 is even**, the median score will Be:

$$\begin{aligned}\text{Median score} &= \frac{(N)^{\text{th}} + (N + 1)^{\text{th}}}{2} \\ &= \frac{(44)^{\text{th}} + (44 + 1)^{\text{th}}}{2} \\ &= \frac{22^{\text{nd}} + 23^{\text{rd}}}{2}\end{aligned}$$

The 22nd member is 30 marks

The 23rd member is 30 marks.

$$\text{Median score} = \frac{30 + 30}{2} = \frac{60}{2} = 30 \text{ mark}$$

Median score 30

Advantages of the median

1. Computation in median is very easy.
2. Median is not affected by extremes of value.
3. It is very easy to understand.
4. It can be obtained by graphic form.
5. The median is easy to determine by mere observation.
6. It does not involve serious calculations

Disadvantages of the median

1. Difficulties come up when large values are involved.
2. The re-arrangement of numbers involves a difficult task
3. It may not be needed for further statistical calculations.
4. It tends to ignore extreme values.

THE MODE

Definition: the mode can be defined as the set most frequently occurring number in a set of numbers or data. It tells us the observation which is most popular. It is the most frequently occurring value in a distribution.

Suitability of mode for use

Mode is suitable for use when we have large array of numbers or want to find the number that appears most in a series of numbers.

The mode may not exist if no item or value repeats itself. Again, mode may not be unique

If more than one item repeats itself and such items have the same highest frequency.

The best and easiest way of calculating the mode of any distribution is to form a frequency table for it.

Example I

The marks scored by economics students in WAEC examinations are as follows: 30, 25, 60, 80, 60, 25, 80, 60, 40, 60, 80, 30, 25. Calculate the mode

Solution

Step I determine the lowest and the highest marks (i.e. 25 and 80).

Step II Arrange the numbers in ascending magnitude (i.e. 25, 30, 40, 60, 80)

Step III prepare a frequency table (table 2.17)

Table 2.17: frequency table of marks scored by economics students in WAEC examination.

Marks	25	30	40	60	80
Frequency	3	2	1	4	3

From the table (table 2.17), the highest frequency is 4 and this corresponds to a mark of 60. The mode is 60

Note: a set of values with two modes is called **bi-modal** but when they are more than two Modes, the set is called **multi-modal**, while a set with only one mode is called **uni-modal**.

Advantages of the mode

1. It is easy to determine.
2. It is easy to understand.
3. It is not affected by extremes of values.
4. When data is not complete, mode cannot be difficult to estimate.
5. It is very easy to compute.

Disadvantages of the mode

1. It is not a very good measure of accuracy.
2. It is irrelevant in further statistical calculation
3. It represents a very poor average.
4. It is difficult to calculate, especially when more than one mode or large numbers are involved.
5. There may be uncertainty in the exact location.
6. Arrangement of data is always tedious.

MEAN DEVIATION

Meaning: the mean deviation is the arithmetic mean of all absolute deviations from the mean. It represents the differences of all the values from the arithmetic mean divided by the number in a given data. The mean deviation is obtained by finding the sum of all the values of

each deviation from the mean (not minding the sign) and then dividing by the numbers (n) of values. The mean deviation is denoted by **M.D.**

Formula for calculating M.D

The formula used for calculating mean deviation is:

$$M.D. = \frac{\sum |x - \bar{x}|}{N}$$

Where N = number of items in the distribution

X = variables

$\bar{x}$ = arithmetic mean

$\sum$ = sum of

Example1

Calculate the mean deviation of the following age of some pupils in Mundra primary school: 4, 5, 6, 8, 10, 3

Solution

Step 1: find the arithmetic mean

$$\begin{aligned} \bar{x} &= \frac{4 + 5 + 6 + 8 + 10 + 3}{6} \\ &= \frac{36}{6} \\ &= 6 \end{aligned}$$

Step 2: calculate the mean deviation using the formula above.

$$\begin{aligned} M.d. &= \frac{|4-6| + |5-6| + |6-6| + |8-6| + |10-6| + |3-6|}{6} \\ &= \frac{2+1+0+2+4+3}{6} \\ &= \frac{12}{6} = 2 \end{aligned}$$

Mean deviation involving frequency

Assume certain figures are given as $x_1, x_2, x_3, \dots, x_n$ with their corresponding frequencies As $f_1, f_2, f_3, \dots, f_n$, the mean deviation for the data is stated below.

$$M.D. = \frac{f_1(x_1 - \bar{x}) + f_2(x_2 - \bar{x}) + f_3(x_3 - \bar{x}) + \dots + f_n(x_n - \bar{x})}{N}$$

Example 2

Calculate the mean deviation for the set of data In table 2.19

Table 2.19: age of NDI students that won scholarship

Age of student	8	10	14	18
Frequency	4	3	5	8

Solution

Step I: prepare a new table (table 2.20) that will reflect all that will be required to make the calculation easy.

Table 2.20: age of ND1 students that won scholarship.

Age (x)	Freq.(f)	fx	x-x	f x-x
8	4	32	5.8	23.3
10	3	30	3.8	11.4
14	5	70	0.2	1.0
18	8	114	4.2	33.6
	$\sum f = 20$	276		$\sum f x-x = 69.2$

Step II: Calculate the arithmetic mean ($\bar{x}$) to enable us compute the fourth column on the Table (table 2.20)

$$\begin{aligned}
 \text{Arithmetic mean} &= \frac{\sum fx}{\sum f} \\
 &= \frac{276}{20} \\
 &= 13.8
 \end{aligned}$$

$$\begin{aligned}
 \text{Mean deviation} &= \frac{\sum f|x-x|}{N} \\
 &= \frac{69.2}{20} \\
 &= 3.46
 \end{aligned}$$

Advantages of mean deviation

1. All the values in a distribution are used for computation.
2. It is used to indicate the average variation of the values in a given data.

Disadvantages of mean' deviation

1. It is difficult to calculate.
2. It ignores the plus and minus signs when calculating.
3. It cannot be used for further mathematical calculation.

VARIANCE AND STANDARD DEVIATION

Meaning: the variance refers to the arithmetic mean of the squares of the deviation of the observation from the true mean. It is also referred to as the “mean square deviation”.

The standard deviation, on the other hand, is the square root of the variance. The standard deviation is also referred to as the “**root mean square deviation**”.

Formulae for calculating variance and standard deviation

The formulae that may be used for calculating variance and standard deviation are stated below.

If for example, there exists observations such as $x_1, x_2, x_3 \dots x_n$, and their true arithmetic Mean is $\bar{x}$, then:

$$(a) \text{ variance} = \frac{\sum (x - \bar{x})^2}{n}$$

$$(b) \text{ standard deviation} = \sqrt{\frac{\sum (x - \bar{x})^2}{n}}$$

Assume that the observations $x_1, x_2, x_3, \dots, x_n$ have frequencies $f_1, f_2, f_3, \dots$. Then the formulae become:

$$(a) \text{ Variance} = \frac{\sum (x - \bar{x})^2}{n}$$

$$(b) \text{ Standard deviation} = \sqrt{\frac{\sum (x - \bar{x})^2}{n}}$$

Example 1

The mark scored by chemistry students in their BAM examination is presented in table 2.20(a). Calculate the variance and standard deviation.

Table 2.21(a); marks scored by economics students in their BAM examination.

Marks	10	20	30	40	50	60
No. of students (frequency)	8	6	12	18	6	4

Solution

Step i: prepare a new table (table 2.21 (b)) to enable us complete all the necessary data.

Table 2.21(b): marks scored by economics students in their BAM examination

Marks (x)	Freq. (f)	Fx	x - $\bar{x}$	x - $\bar{x}$ ²	f x - $\bar{x}$ ²
10	8	80	23.7	561.69	4493.52
20	6	120	13.7	187.69	1126.14
30	12	360	3.7	13.69	164.28
40	18	720	6.3	39.69	714.42
50	6	300	16.3	265.69	1594.14
60	4	240	26.3	691.69	2766.76
	fx = 54	$\sum fx = 1820$			10,859.26

Step 11: calculate the arithmetic mean ($\bar{x}$) to enable us compute for column four on the table above.

$$\begin{aligned} \bar{X} &= \frac{\sum fx}{\sum f} \\ &= \frac{1820}{54} = 33.7 \end{aligned}$$

Arithmetic mean ($\bar{x}$) = 33.7

From the data available from table 2.21 (b)

$$(a) \text{ Variance} = \frac{\sum f |x - \bar{x}|^2}{\sum f}$$

$$\begin{aligned}
&= \frac{10,859.26}{54} \\
&= 201.1
\end{aligned}$$

$$\begin{aligned}
\text{(b) Standard deviation} &= \frac{\sqrt{\sum |x-x|^2}}{\sum f} \\
&= \sqrt{201.1} \\
&= 14.1
\end{aligned}$$

Advantages of standard deviation

1. It represents the best measure of dispersion.
2. It makes use of every value in the distribution.
3. It takes mathematical signs into consideration.
4. It is very useful in further mathematical analysis.
5. It is also useful in sample theory.

Disadvantages of standard deviation

1. It lays more emphasis on the use of extreme values.
2. Its calculation is very difficult and tedious.

PRODUCTION

Production is defined as the various economic activities aimed at the creation of goods and services and the distribution of these to the final consumers for the satisfaction of human wants.

Production can also be defined as the creation of utility. All goods and services produced must possess utility, which means that they must be capable of satisfying certain human wants. Production include the creation of tangible goods such as clocks, cars, tables, TV sets, houses, books etc. as well as services rendered by some professionals such as doctors, lawyers, hair dressers, soldiers, police, teachers, and mechanics etc.

It should be noted that production is said to be complete when the goods and services produced gets to the final users or consumers. The production process involves three major groups of people. These are producers, distributors and consumers.

TYPES OF GOODS

Goods and services can be classified into two major categories. These are consumer goods and producers or capital goods.

- a) **Consumer goods:** these are goods and services that can satisfy the consumer's immediate needs. These goods do not required further process of production for their use by the consumers. examples of consumer goods includes motor cars, milk, cake, radio, television set, the service of a soldier, police, hair dresser, barber, nurse, lawyer, etc.

Consumer goods can further be grouped into two classes. These are durable and non-durable goods.

- **Durable goods:** durable goods are goods that can be used over and over again. They are used many times or for many years before they get worn out. Examples includes radio, television set, shoes, tables, shirt, cooking pots, spoons etc. they are generally referred to as consumer durable goods.

- **Non-durable goods:** these are goods that are used up from one use. In other words, they are used up at once. Examples include meat, milk, bread, drugs, etc.
- b) **Producer or capital goods:** these are goods used by people to produce some other commodities or services. Examples includes buildings, motor cars, Lorries, tailor's tools, machines etc. these goods are used to carry out productive activities.

TYPES OF PRODUCTION

Production is grouped into two major categories. These are direct and indirect production.

- a) **Direct production:** direct production is the type of production in which an individual produces goods and services only for family use or consumption. The goods and services so produced are not for sale; rather they are to be used by family. It is usually very small in size and family labor is usually employed.
- b) **Indirect production:** indirect production is the type of production in which goods and services are produced in large scale, mainly for sales or exchange for other needs. This type of production involves the use of modern equipment and skilled labor to be able to make surplus goods and services that they producer will use to satisfy other wants. It is practically impossible for a single producer to satisfy all his wants without depending on others for exchange of goods and services.

Indirect production is sub-divided into three major groups. These are primary, secondary and tertiary production.

- **Primary production:** primary production refers to the extraction of raw materials provided by nature. It is concerned with the process of obtaining raw materials or resources in their natural form from the land, air and water. Primary production forms the basis for further production. Examples of primary production include agriculture, mining, fishing, lumbering, etc.
- **Secondary production:** secondary production involves the transformation of or conversion of basic raw materials or semi-final forms that are acceptable to the consumers. Secondary production embraces all forms of manufacturing and construction of. The raw materials or resources obtained from the extractive sector are transformed into finished products such as processed food, houses, roads, clothes, cars, etc.
- **Tertiary production:** tertiary production is concerned with the provision of commercial and professional services to the people. The goods so produced at the primary and secondary production levels are distributed to the people for consumption. The people involved in this aspect of production include those in the commercial service like wholesalers, retailers, transporters etc.as well as those involved in rendering professional services like soldiers, police, doctors, lawyers, musicians, hair dressers, teachers etc.

FACTORS OF PRODUCTION

Factors of production refers to agents, components or resources which are combined together to produce goods and services. There are four factors of production. These are:

- a) Land
- b) Labour
- c) Capital
- d) Entrepreneur

- ❖ **Land:** land is defined in economics as a free gift of nature. Land in economics does not only include land surface of the earth but all other free gift of nature or natural resources like forest, mineral resources, rivers, oceans, atmosphere, etc. unlike other factors of production, the supply of land is limited. The reward for land is **rent**.

Characteristics or features of land

- Land is immobile
- Land is a free gift of nature
- The supply of land is fixed
- Land is subject to diminishing returns
- Variability
- Rent
- Land has no cost of production
- Land is heterogeneous

Law of diminishing return:

The law of diminishing return states that as successive units of variable factor (e.g. labor or capital), is applied to a given fixed factor (e.g. land), output will increase at first but it will get to a point at which the addition of one more unit of the variable factor will result in less additional units of output. The law of diminishing returns, sometimes called the **law of variable proportions**, is applicable to both the agricultural and industrial sectors that use both the fixed and variable factors of production.

- ❖ **Labor:** labor as a factor of production is defined as all forms of human efforts put into or utilized in production. It also refers to man's mental and physical exertions generated in the process of production.

Human beings provide the necessary labor which combines with other factors to provide goods and services. The reward for labor as a factor of production comes in form of **wages and salaries**.

Types of labour

- **Unskilled labor:** this category requires little or no formal education. They do not use mental effort; rather they make use of their physical efforts or energy in production. Their jobs are popularly referred to as a **brown collar jobs**.
- **Skilled labor:** this category makes use of their mental efforts in productive activities. This labor has undergone a relatively long and specialized type of training in institutions of higher learning. These categories of workers are also referred to as a **white collar jobs**.

Characteristics of labor

- labor has feelings
 - Labor is mobile.
 - labor is skillful
 - labor requires motivation
 - Labor is perishable
 - Labor is a human factor
- ❖ **Capital:** capital can be defined as a man-made assets used in production. In other words, capital refers to man-made wealth or goods used to produce service. It may also be defined as the stock of previous wealth invested in other to produce future wealth.

Capital, when properly combined with other factors, produces goods and services. Examples of capital includes physical cash, cutlass, hoe, machines, buildings, motor vehicles, raw materials, semi-finished goods tools and other equipment used in the production of goods and services. The reward for capital is **interest**.

Characteristics of capital

- Capital is man-made
- Capital is durable
- Capital exist in different forms
- Capital ensures large scale production
- Capital subject to depreciation
- It promotes division of labor

Types of capital

- **Fixed capital:** These are assets that are not used up in the course of production. Fixed assets include those durable assets of a business that can last for a very long time.
 - **Circulatory or working capital:** these are assets which are used up in the course of production. These consist of capital goods which either change their form or are used up in the process of production.
 - **Current or liquid capital:** Current capitals are the type of capital that is required for the day-to-day running of productive activities. These are also changed from one form to another.
 - **Social capital:** this includes those forms of capital or assets provided by the government that aid production. Examples are amenities provided by government such as roads, electricity, water, telephones etc.
- ❖ **ENTREPRENEUR:** an entrepreneur can be defined as the factor of production that co-ordinates and organizes other factors of production (land, labor and capital) in order to produce goods and services. The entrepreneur bears all the risks and takes major decisions of the business. The reward for entrepreneur is **Profit**.

Characteristics of entrepreneur

- Risks bearer
- Organization
- Decision making
- Controls other factors of production

PRODUCTION POSSIBILITY CURVE (PPC):

The production possibility curve (PPC) also known as the production possibility boundary (PPB), refers to a graph or a curve showing the possible combinations of different commodities that can be produced in a given economy, given the prevailing level of technology, if all the available productive resources are efficiently utilized. In other words, PPC is a graphical illustration of all the possible combinations of two or more types of commodities which a society can produce using a given quantity of resources.

The idea behind the production possibility curve (PPC) is that in order to produce a particular commodity, the production of another commodity needs to be sacrificed.

RELATIONSHIPS BETWEEN PRODUCTION POSSIBILITY CURVE AND OPPORTUNITY COST

Opportunity cost can be defined as an expression of cost in terms of forgone alternatives. It is the satisfaction of one's want at the expense of another want. It refers to the wants that are left unsatisfied in order to satisfy another more pressing need. The production possibility curve (PPC) is directly connected with opportunity. The PPC involves sacrifice in the production of one commodity in order that another one can be produced.

Once more resources are allocated to the production of a commodity, fewer resources would be allocated to the production of others (because resources are scarce).

CONCEPTS OF TOTAL, AVERAGE AND MARGINAL PRODUCTIVITY

❖ **Total product (TP):** total product is defined as the total quantity of commodities produced at a particular time as a result of the combination of all the factors of production.

Total production is expressed mathematically as: **TP = AP x Labour**

Example:

If 30 men were employed in a farm and they produce an average of 15 tonnes of cassava per person, calculate the total product.

Solution:

$$TP = AP \times \text{Labour}$$

$$\begin{aligned} TP &= 15 \text{ tonnes} \times 30 \\ &= 450 \text{ tonnes} \end{aligned}$$

❖ **Average product (AP):** average product (AP) is defined as the output per unit of the available factor (labour or capital) employed. This is obtained by dividing the total output by the number of labour or capital employed.

$$AP \text{ can be expressed mathematically as: } AP = \frac{\text{total output}}{\text{No. of labour (capital) employed}}$$

Example:

If 3,000 tonnes of cassava were harvested by 60 men in a farm on daily basis, calculate average output.

Solution:

$$\begin{aligned} AP &= \frac{\text{total output}}{\text{No. of labour (men)}} \\ &= \frac{3,000 \text{ tonnes}}{60 \text{ men}} \\ &= 50 \text{ tonnes / person} \end{aligned}$$

❖ **Marginal product:** the marginal product (MP) may be defined as the addition to the total output brought about as a result of the employment of an additional unit of a variable factor. MP can be expressed mathematically as:

$$MP = \frac{\text{Change in TP}}{\text{Change in variable factor}}$$

Example:

If 3,080 tonnes of cassava were harvested from the same farm as a result of an additional man to the 60 men, calculate the marginal product.

Solution:

$$MP = \frac{\text{Change in TP}}{\text{Change in variable factor}}$$

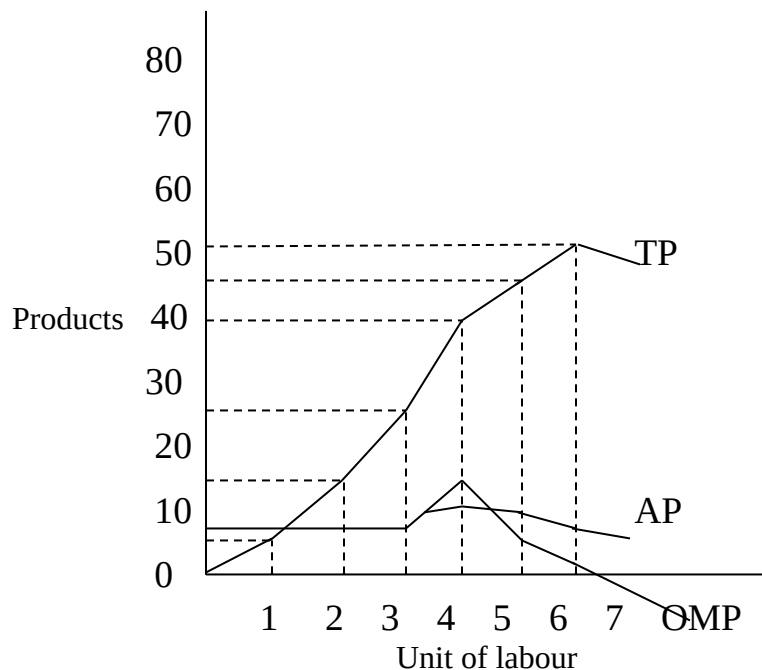
$$MP = \frac{(3,080 - 3,000)}{(61 - 60)}$$

$$= \frac{80}{1}$$

$$MP = 80 \text{ tonnes}$$

RELATIONSHIPS BETWEEN TOTAL PRODUCT, AVERAGE PRODUCT AND MARGINAL PRODUCT

The relationship between total product, average product and marginal product can best be demonstrated by a graph. TP, AP, and MP rise initially. TP rises sharply and curves at a maximum when MP is zero. TP declines after MP assumes negative value.

**POPULATION**

Population is defined as the total number of people living within a country or a geographical area at a particular time. Population, in other words, refers to the total number of children, adults, (men and women), youth (boys and girls) living in a given geographical area, which may be a town, village, or country at a specific time.

The study of population is very important for many reasons. Such reasons include the determination of the working population, allocation of resources, the availability of labour, etc.

POPULATION CENSUS

Population census may be defined as the head count of all nationals of a country at a particular time. It refers to the counting by government of all the children, girls, men and women, including the disabled, in a country at a given period of time.

Censuses are usually taken every 10 years. When series of censuses has been undertaken properly, it becomes easier, using the rate of growth, to estimate the population between the periods of counts. Population census provides information about age, sex, occupation, residence, etc.

CHARACTERISTICS OF GOOD POPULATION CENSUS

- a) It must involve the physical counting of people rather than by proxy.
- b) It must be conducted by government of the state or country where the census is taking place.
- c) It must involve experts who are knowledgeable in population studies and they must be of high integrity.
- d) It must be conducted at the same time (simultaneously) throughout the country.
- e) It must reveal the population of a country at a specific period of time.

TYPES OF POPULATION CENSUS

1. **Defacto population census:** this is the type of population census which involves the counting only those who are present physically during census. In this group, only those that are present physically or seen are counted.
2. **De jure population census:** this is the type of population census which involves the counting of people who have been permanent residents of a specific area. It does not matter whether the person is present or not. This is the population census commonly referred to as counting by proxy.

AGRICULTURE

Agriculture is simply defined as the production of crops and rearing of animals for man's use. In other words, agriculture is the deliberate effort made by man to till the soil, cultivate crops and rear animals for food and other purpose.

Agriculture is one of the most important sectors of the economy of many West African countries. Even though agriculture is practiced widely at the subsistence level, it employs about 60-70 percent of the population of West African countries.

SYSTEMS OF AGRICULTURE

1. **Peasant farming:** peasant farming, also known as subsistence farming, is defined as the type of agriculture which is concerned with the production of food for the farmer and his family. In other words, it is a system of farming whereby a farmer cultivates crops and rears animals in order to produce food for use by himself and his family only.
It requires a small area of land, family labour is usually employed, there is usually little or none left for sale and also it requires small capital to operate.
2. **Plantation farming:** plantation farming involves the cultivation of some economic crops in large areas of land. In this type of farming only one crop (monoculture) is grown in a large area of land.
It requires a large area of land, only one crop is cultivated, it requires huge capital investment, and also harvest or yield is usually great.
3. **Mechanized farming:** this also referred to as commercial agriculture, is the type of agriculture which is concerned with the production of food, animals and cash crops in large quantities mainly for sale.

This requires special skills and technical education or labour, special marketing strategies, and also it requires a large area of land.

4. **Co-operative farming:** it is the types of farming in which farmers come together to form a union for the purpose of farming together to enable them enjoy some incentives from government. Such incentives may include loans, input supply like fertilizers, subsidies and easy disposal of their products through organized marketing.

It has organized market for their products, output is high because of the resources put together by the farmers, and also it is a union of farmers.

IMPORTANCE OR CONTRIBUTION OF AGRICULTURE TO ECONOMIC DEVELOPMENT OF WEST AFRICA

1. Development of towns.
2. Source of foreign exchange.
3. Generation of income.
4. Provision of food.
5. Employment for the working population.
6. Provision of materials for clothing.
7. Provision of materials for shelter.
8. Provision of raw materials for industries.
9. Provision of market for industrial goods.

MONEY

Money is anything that is generally acceptable as a medium of exchange and in the settlement of debts. Money can also be defined as anything that is generally acceptable as means of payment.

QUALITIES OF MONEY

1. General acceptability
2. Portability
3. Homogeneity
4. Stability
5. durability
6. Relative scarcity
7. No intrinsic value
8. Recognisability
9. Divisibility

FUNCTIONS OF MONEY

1. Standard of deferred payment
2. Store value
3. Unit of account
4. Medium of exchange
5. As a measure of value

INFLATION AND DEFLATION

INFLATION:

This may be defined as a persistent rise in the general price level of goods and services. Inflation occurs when the volume of purchases is permanently running ahead of production and too much money in circulation chasing too few goods.

Types of Inflation

1. **Cost-push inflation:** this type of inflation occurs when increases in cost of production are passed on to consumers in the form of high prices for the goods and services on sale. The prices of goods and services are pushed up by rising costs.
2. **Hyper-inflation:** this is also known as galloping or run-away inflation, it occurs when persistence inflation becomes uncontrollable and the value of money keeps declining rapidly.
3. **Demand-pull inflation:** this occurs when consumers have high purchasing power leading to increase in aggregate demand without a corresponding increase in supply. In other words, this type of inflation occurs when the demand for goods and services is greater than their supply.
4. **Persistent or creeping inflation:** persistent or creeping inflation, also known as chronic inflation, occurs when there is a slow but steady rise in the volume of purchasing power and a fall in supply of goods and services. In other words, when inflation involves a slow but steady rise in the general prices of goods and services, it is known as creeping inflation.

Causes of inflation

1. War
2. Low production
3. Increase in demand
4. High cost of production
5. Excessive bank lending
6. Population increase
7. Budget deficit
8. Increase in salaries and wages
9. Hoarding
10. Money laundering

Effect of inflation

1. Higher tax yield
2. The negative effect of inflation
3. Higher output
4. The positive effects of inflation
5. It discourages savings
6. Creditors lose
7. Loss of value money
8. Fall in standard of living

Control of inflation

1. Use of fiscal measure
2. Industrialization
3. Increased production

4. Effective price control system
5. Discouragement of importation
6. Use of income policies
7. Checking the activities of hoarders
8. Granting of subsidy to enterprises

DEFLATION

Deflation may be defined as a continuous fall in the price level of goods and services as a result of decrease in the volume of money in circulation. Since prices fall, the value of money rises during deflation. A given sum of money can purchase more goods and services. It should be noted that deflation is the opposite of inflation.

Causes of deflation

1. Budget surplus
2. Increase in bank rate
3. Increase in production
4. Increase in taxation

Effect of deflation

1. It results in unemployment
2. Reduction in investment
3. Creditors gain
4. Decline in profit
5. Fall in prices of goods
6. It encourages export
7. It discourages import
8. It encourages savings
9. Fixed income earners gain

Control of deflation

1. Reduction in bank rate
2. Use of deficit budget
3. Reduction in taxation
4. Increase in wages and salaries
5. Use of open market operation

INDUSTRIALIZATION

A firm is the basic unit within which factors of production are organized for the purpose of producing wealth. A firm is an entity which specializes in the production and distribution of goods under one administration or management.

An industry, on the other hand, is defined as a group of firms producing similar products and under separate administration or management. In other words, an industry includes a collection of different firms which produce similar goods under separate management.

TYPES OF INDUSTRY

1. **Manufacturing industry:** this operates at the secondary production level. These are industries which can convert raw materials to produce finished goods, using special machines and equipment.
2. **Mining industries:** this industry is concerned with the extraction of mineral resources from the soil by special devices. Mining is a primary production level.

3. **Construction industries:** This industry is concerned with the assembling of different components in order to get a finished product.
4. **Transport industry:** this is concerned with the movement of passengers and goods from one place to another. The movement of passengers or goods can be done by land, sea, and air.
5. **Processing industry:** this industry is concerned with the conversion of some raw materials into finished goods (products) that can be used directly to satisfy human wants.
6. **Power and energy industry:** this industry is concerned with the generation, transmission and distribution of electricity.

LOCATION OF INDUSTRY

Location of industry may be defined simply as the siting or establishment of a firm or industry in a particular place. An industry may be established either by individuals or government, either for economic or political reasons.

FACTORS INFLUENCING THE LOCATION OF AN INDUSTRY

1. Availability of capital
2. Nearness to source of power
3. Availability of labour
4. Nearness to market
5. Proximity to source of raw materials.
6. Adequate transport network
7. Political stability
8. Favorable climate
9. Government policies

DEMAND AND SUPPLY

DEMAND

Demand may be defined as the ability and willingness to buy a specific quantity of goods or services at a given price and at a particular period of time.

In economic, demand is quite different from want or need, want or need refers to a mere desire for a commodity but not backed up by the willingness and ability to pay for it.

Law of demand

The law of demand states that all things being equal, the higher the price, the lower the quantity of goods that will be demanded; or the lower the price, the higher the quantity of goods that will be demanded. This law is regarded as the first law of demand and supply.

Demand schedule

Demand schedule can be defined as a table showing the relationship between prices and the quantity of that commodity demanded. In other words, a demand schedule is a table which shows the different quantities of a commodity that would be bought at various prices, at a particular time. There are two types of demand schedule namely: individual demand schedule and market demand schedule.

Individual demand schedule:

Price per tin milk	Quantity demanded (no. of tins per week)
100	10
80	20
60	30
40	40
20	50

Market demand schedule

Price Per Tin (₦)	Quantity demanded by				Total quantity purchased per week
	Mal. Musty	Mr. David	Mrs. Maryam	Mr. Johnson	
100	10	15	30	20	75
80	20	25	40	40	125
60	30	35	50	60	175
40	40	45	60	70	215
20	50	55	70	80	255

Demand curve

This is a graph showing the relationship between the price and quantity of a commodity demanded. In other words, a demand curve can be defined as a graphic or diagrammatic representation of a demand schedule. It should be noted that demand curve is derived from demand schedule.

Types of demand

1. **Competitive demand:** this occurs when two commodities are fairly closed substitute to each other. In other words, they serve the same purpose or perform a similar function such that an increase in the demand for one will result in a fall in demand for the other.
2. **Derived demand:** this type of demand which occurs as a result of demand for other commodities. The demand for one commodity will necessitate the demand for another.
3. **Joint or complementary demand:** this is a demand which occurs when two commodities that are related to each other are demanded at the same time.
4. **Composite demand:** this occurs when a commodity is required to serve two or more purposes.

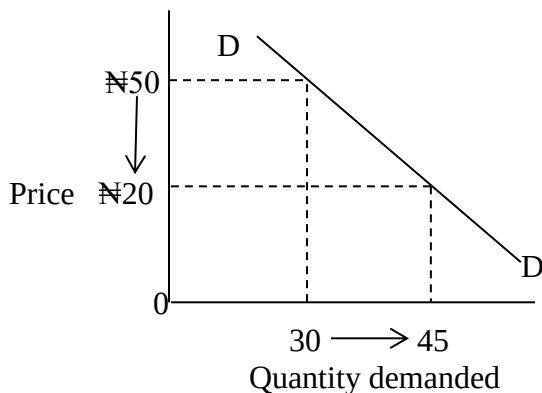
Factors affecting demand

1. Price
2. Income of the consumer
3. Population
4. The price of other commodities
5. Change in taste of consumer
6. Periods of festival
7. Change in fashion
8. Taxation
9. Expectation of changes in price
10. Advertisement

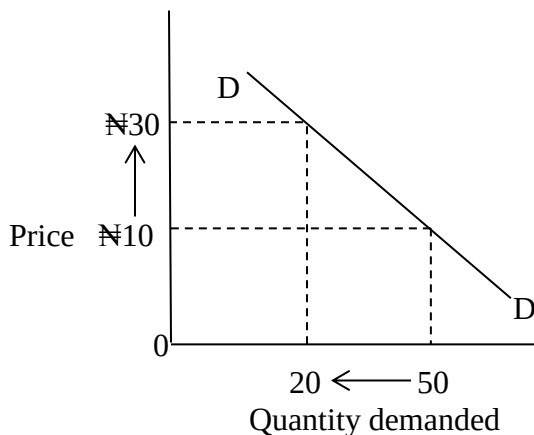
Change in quantity demanded

A change in quantity demanded of a commodity means a movement from one point to another on a demand curve. The cause of the change in quantity demanded is due to change in the price of commodity under consideration. The quantity of commodity demanded changes with price. A change in quantity demanded is of two parts:

- **Increase in the quantity demanded:** there is an increase in the quantity demanded provided the quantity purchased increases as a result of a decrease in the price of the commodity.



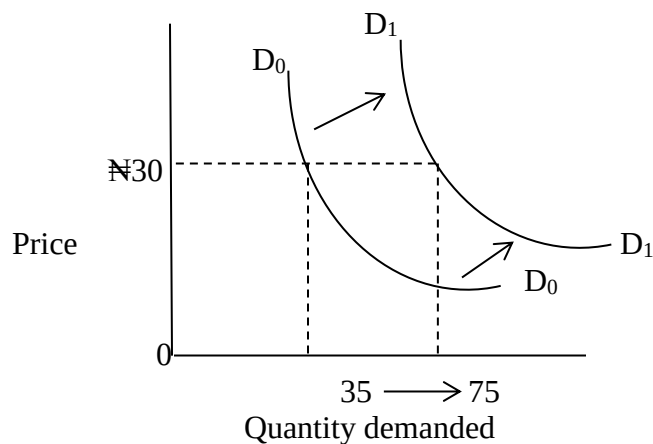
- **Decrease in the quantity demanded:** in this case, there is a decrease in the quantity demanded if the quantity of the commodity purchased decreases as a result of an increase in price.



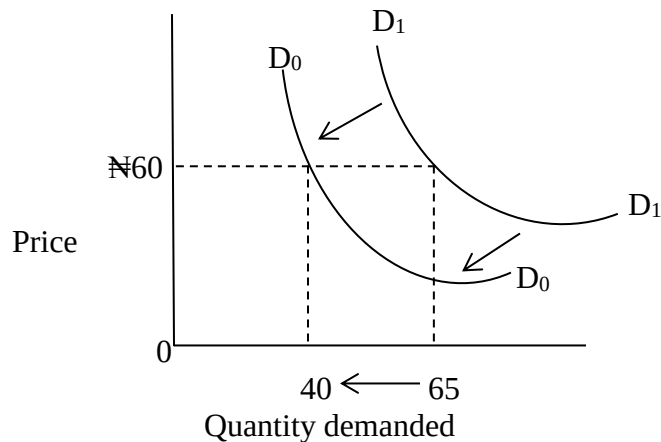
Shift or change in quantity demanded

A shift or change in demand in economics is quite different from change in the quantity demanded. There is change in demand if the demand curve shifts to an entirely new position. In this case there is a new demand schedule and demand curve showing that the old price more or less of the commodity would be purchased. A shift or change in demand is determined by other factors affecting demand except the price of the commodity.

A shift in demand is also grouped into two divisions namely; increase in demand and decrease in demand.



Rightward shift (increase) in demand



Leftward shift (decrease) in demand

Elasticity of demand

Elasticity of demand may be defined as the degree of responsiveness of demand to little change in the price of a commodity.

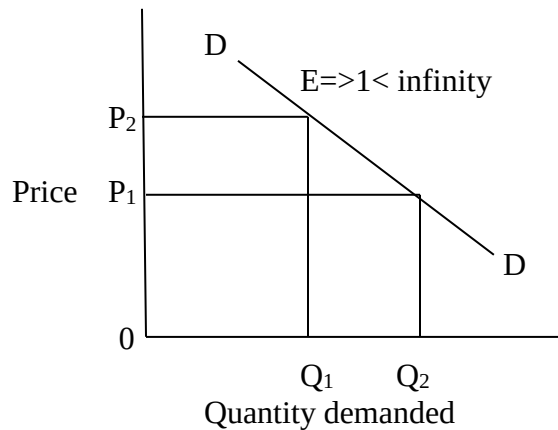
Elasticity of demand measures the extent to which the quantity of a commodity demanded by consumers changes as a result of a little change in price of the commodity.

Price elasticity of demand

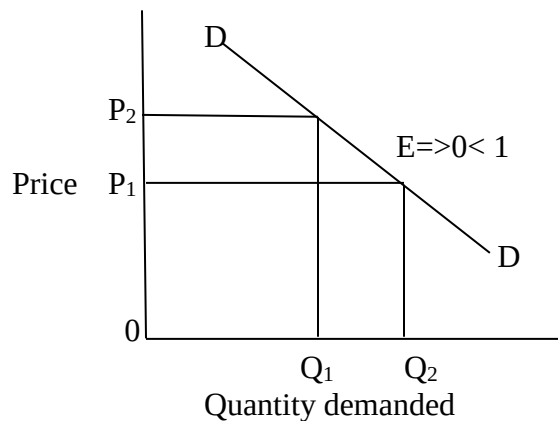
Price elasticity of demand, also known as the co-efficient of price elasticity, may be defined as the degree of responsiveness of demand to little changes in price of goods and services. In other words, it is the degree of responsiveness of quantity demanded to a small change in the price of the commodity.

Types of price elasticity of demand

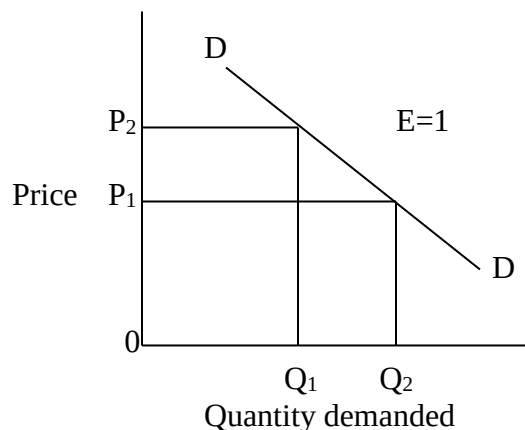
1. **Elastic demand:** demand is said to be elastic if a small change in price leads to a greater change in the quantity of goods demanded. In this case elasticity is greater than one or unitary, i.e. $E > 1 < \text{infinity}$.



2. **Inelasticity of demand:** demand is said to be inelastic if a larger change in price leads to a small or slight change in quantity of goods demanded. In this case, elasticity is less than one but greater than zero, i.e. $E > 0 < 1$.

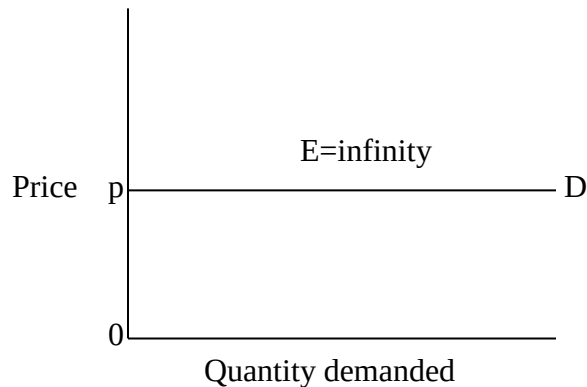


3. **Unity or unitary elasticity of demand:** demand is said to be unitary when a change in price leads to an equal change in quantity of goods demanded. In other words, 5% change in price will lead to a 5% change in demand. In this case elasticity is equal to one, $E = 1$.

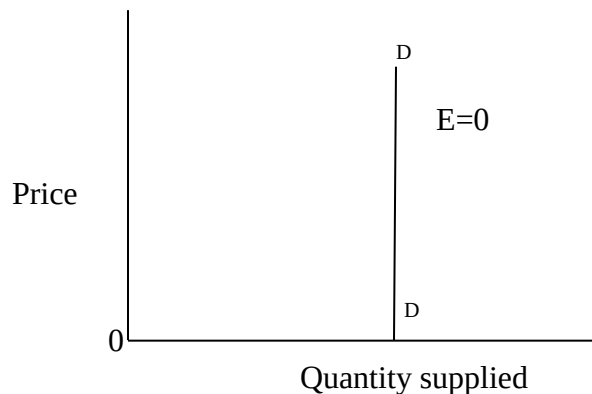


4. **Perfectly elastic demand or infinitely elastic demand:** demand is said to be perfectly elastic when a change in price brings about an infinite effect on the quantity of goods

demanded. In other words, the same quantity of goods is demanded irrespective of changes in price. In this case elasticity is equal to infinity, i.e. $E = \text{infinity}$.



5. **Perfectly elastic demand or zero elastic demand:** demand is said to be perfectly elastic if change in price has no effect, whatsoever on the quantity of goods demanded. In this case, elasticity is equal to zero, i.e. $E = 0$



Measurement of Elasticity of Demand

Example:

The price of bread in 2023 was increased from ₦40 to ₦50 and the quantity bought per week by a consumer decreased from 160 loaves to 80.

- Present the above data in a table
- Calculate the co-efficient of price elasticity of demand.
- What type of elasticity is this and how did you arrive at your conclusion.

Solution:

(i)

Price ₦	Quantity demanded
40	160
50	80

(i) Percentage change in quantity demanded

$$= \frac{\text{change in QD}}{\text{Old QD}} \times \frac{100}{1}$$

$$= \frac{(160 - 80)}{160} \times \frac{100}{1}$$

$$= \frac{80}{160} \times \frac{100}{1}$$

$$= 50\%$$

(ii) Percentage change in quantity demanded

$$= \frac{\text{change in price}}{\text{Old price}} \times \frac{100}{1}$$

$$= \frac{(\text{N}50 - \text{N}40)}{\text{N}40} \times \frac{100}{1}$$

$$= \frac{10}{40} \times \frac{100}{1}$$

$$= 25\%$$

Co-efficient of price elasticity of demand

$$ED = \frac{50\%}{25\%} = 2$$

(ii) The co-efficient of elasticity of demand is elastic. It is elastic because elasticity is greater than 1.

Income elasticity of demand

Income elasticity of demand is defined as the degree of responsiveness of demand to changes in income of consumers. In other words, it measures how changes in income of consumers will affect the quantity of commodities demanded by such consumers. It should be noted that income elasticity of demand is negative to inferior goods since an increase in income will lead to a decreased demand for them.

Types of income elasticity of demand

- i. **Positive income elasticity of demand:** this is the type of income elasticity of demand in which an increase in income of consumers will equally leads to an increase in the quantity of commodity demanded. This is applicable mainly to normal commodities.
- ii. **Negative income elasticity of demand:** this is the type of income elasticity of demand in which an increase in the quantity of commodities demanded. In this case, as income increase, demand for commodities falls. This is applicable to inferior goods.

Measurement of income elasticity of demand

Example:

A weekly income of a clerk was increased from ₦100 to ₦125 as a result of his promotion in the office. He is able to purchase 300 loaves of bread instead of 200 per week.

- (i) Calculate the co-efficient of his income elasticity of demand
- (ii) Is the demand elastic or inelastic? Why?
- (iii) What kind of goods is bread to the consumer?

Solution:

Income		Quantity demanded	
Old	New	Old	New
		(Loaves)	(loaves)
100	125	200	300

1. (a) percentage change in quantity demanded

$$= \frac{\text{new old QD} - \text{old QD}}{\text{Old QD}}$$

$$= \frac{300-200}{200} \times \frac{100}{1}$$

$$= \frac{100}{200} \times \frac{100}{1}$$

$$= 50\%$$

- (b) Percentage changes in income

$$\frac{\text{New income} - \text{old income}}{\text{Old income}} \times \frac{100}{1}$$

$$\frac{125 - 100}{100} \times \frac{100}{1}$$

$$\frac{25}{100} \times \frac{100}{1} = 25\%$$

$$(c) \frac{\% \Delta Qd}{\% \Delta \text{income}} = \frac{50\%}{25\%}$$

$$= 2.0$$

The co-efficient of income elasticity = 2

2. The co-efficient of income elasticity of demand is elastic. It is elastic because elasticity is greater than 1.
3. The kind of goods bread is to the consumer is normal goods because as the income increases, his demand for bread too also increases thus indicating a positive type of income elasticity of demand.

Cross elasticity of demand

Cross elasticity of demand may be defined as the degree of responsiveness of demand for a commodity to changes in the price of another commodity. In other words, cross elasticity of demand refers to the proportionate change in the quantity of goods (X) demanded over the proportionate change in the price of another goods (Y) demanded. Cross elasticity of demand is applicable mainly to goods that are close substitutes as well as complementary goods.

Factors affecting elasticity of demand

1. The income of the consumer
2. Nature of commodity
3. Number of uses to which a commodity is put
4. Close substitute of a commodity
5. One's habit

Importance of elasticity of demand

1. Imposition of taxes by government
2. Increase in revenue
3. Determination of maximum output
4. Determination of cross elasticity

SUPPLY

Supply may be defined as the quantity of a commodity which a producer is willing and able to offer for sale at a particular price, and at a particular time. In other words, the supply of a commodity is the quantity of that commodity which a producer is willing and able to sell at a given price over a period of time.

The law of supply

The law of supply states that, all things being equal, the higher the price the higher the quantity of a commodity that will be supplied or the lower the price the lower the quantity of the commodity that will be supplied.

This law is often refers to as the second law of demand and suppl. This law explains that when the price of commodity is high in the market, more quantity of it will be supplied by the producer and vice versa.

Supply schedule

Supply schedule can be defined as a table showing the relationship between price and the quantity of that commodity supplied. In other words, supply schedule is a table which shows the different quantities of a commodity which would be supplied at various prices and at a particular time.

Farmer's supply schedule for bags of rice:

Price per bag	Quantity supplied (no. of bags of rice)
100	50
80	40
60	30
40	20
20	10

A Market supply schedule for bags of rice

Price Per bag (₹)	Quantity supplied by				Total quantity supplied
	Mal. Musty	Mr. David	Mrs. Maryam	Mr. Johnson	
100	10	15	30	20	75
80	20	25	40	40	125
60	30	35	50	60	175
40	40	45	60	70	215
20	50	55	70	80	255

Types of supply schedule

1. **Individual supply schedule:** is a table which shows the different quantities of a commodity which a producer offers for sale at various prices and at a particular time.
2. **Market supply schedule:** is a schedule of all producers or suppliers of a commodity in a market. In other words, a market supply schedule is a table which shows the total quantity of a commodity which all producers of that commodity are willing and able to supply at various prices, at a particular period of time.

Supply curve

Supply curve is a graph showing the relationship between price and quantity of that commodity supplied. In other words, a supply curve can be defined as a graphic or diagrammatic representation of a supply schedule.

THE MARKET

A market is a point of contact, place or any means of communication whereby sellers and buyers can communicate with one another, to exchange goods and services at prices determined by the market forces.

Types of Markets

Markets can be grouped on the basis of the type of commodities purchased and sold or on the basis of the channel of movement of finished products from the producers to consumers, or on the basis of prices.

Types of market according to commodities sold in them:

Money market: Money market is the type of market for short term loan. It consists of institutions or individuals who either have money to lend or wish to borrow on a short term basis.

Capital market: Capital market is a market for medium term and long term loans. It serves the needs of industries and the commercial sector. It comprises all institutions which are concerned with either the supply of or demand for long term loan.

Consumer goods market: Consumer goods market is the type in which finished products ready for use by consumers are sold and bought.

Primary products market: This is the type of market in which primary products in their raw forms are sold and bought.

Factor market: This is the type of market in which the factors of production are sold and bought.

Foreign exchange market: This is a market which deals with foreign exchange transactions and it involves the buying and selling of foreign and it involves the buying and selling of foreign currencies.

Labour market: Labour market is the type in which workers and employers are in close contact for the purpose of rendering services.

Stock exchange market: This is a market where investors can buy and sell existing securities like shares, stocks, debentures, etc. This is a market where those who are interested in the purchase of securities are brought into contact with the sellers.

Types of market according to channel of distribution:

(1) **Wholesale market:** This is the type of market in which a trader called wholesaler buys commodities in large quantities from the manufacturer and sells in small quantities to the retailer.

(2) **Retail market:** This is the type of market in which a trader known as a retailer buys commodities from the wholesaler and sells in bits to the final consumers.

(3) **Types of market according to prices:** The types of market based on the prices of commodities are grouped into two: perfect market and imperfect market.

Perfect Market

A perfect market may be defined as one in which buyers or sellers cannot influence the prices of goods and services. Perfect market is also called competitive market or perfect competition.

Conditions Necessary for a Perfect Market

- (1) **Homogeneous goods:** The goods bought and sold in a perfect market must be homogeneous; that is, they must be identical. They must be of the same size, shape, weight, Color, etc. The goods must be the same in the eyes of the consumer.
- (2) **Free entry and exit:** There is free entry into the market for new firms or persons and free exit for the existing ones.
- (3) **Large buyers and sellers:** There is a large number of buyers and sellers, each of whom has no control over the prevailing prices.
- (4) **No preferential treatment:** In perfect market all buyers must be treated equally. Also, sellers must not sell either below or above prevailing prices or show favoritism.
- (5) **Perfect knowledge:** There must be a perfect information or knowledge about the price of the goods or services.
- (6) **Common price:** In a perfect market, the commodity concerned bears the same price tag throughout the market.
- (7) **Portable goods:** The goods to be sold must be easy to carry from one place to another; for where such goods are not portable, they will be sold at different prices.
- (8) **Easy transfer of factors of production:** In perfect market, factors of production are transferred easily to where they are needed.

Imperfect Market

An imperfect market may be defined as the market in which prices of goods or services can easily be influenced by the sellers or buyers. Imperfect market is also called imperfect competition.

Types of Imperfect Market

- (1) **Monopoly:** This is an imperfect market in which there is a single seller of a particular good or service.
- (2) **Monopolistic competition:** This is a market where there is a large number of producers dealing in different products or services such that no product of one firm is seen as a perfect substitute for that of another.
- (3) **Duopoly:** This is an imperfect market in which there are only two producers of the same commodity.
- (4) **Oligopoly:** This is an imperfect market in which there are few producers or sellers of the same commodity.
- (5) **Monopsony:** This is the type of market in which there is only one buyer for a product.